A Hybrid Network With Structural Constraints for SAR Image Scene Classification

Xiaoxue Qian, Fang Liu, *Senior Member, IEEE*, Licheng Jiao, *Fellow, IEEE*, Xiangrong Zhang, *Senior Member, IEEE*, Puhua Chen, *Member, IEEE*, Lingling Li, *Member, IEEE*, Jing Gu, *Member, IEEE*, and Yuanhao Cui

***Abstract*— Data-based image classification methods, such as convolutional neural networks (CNNs), have achieved state-of-the-art performance. They usually leverage thousands of labeled samples to train the networks but ignore some prior knowledge. However, labeled samples are difficult to be obtained for synthetic aperture radar (SAR) images. Model-based methods are adept at utilizing the prior information of data, while they have to introduce some restrictions or assumptions during the realization of models. Consequently, to develop the advantages of both methods and improve their disadvantages, we propose a hybrid network by coupling the data-based with model-based methods for SAR image scene classification in this article. First, to fully use the prior information of SAR images and large amounts of unlabeled samples, we improve the G^0-based variational Bayesian inference model (GVBI) and construct a G^0-based convolutional variational auto-encoder (GCVAE) for unsupervised learning of the distributional characteristics of SAR images. After that, we further extend the GCVAE by combining it with CNN, resulting in a stronger hybrid network to classify SAR images with a few labeled samples. In addition, considering the abundant structural information is crucial for SAR image classification, we design a sketch fitter and two structural constraints on both pixel and sketch spaces to assist the hybrid network to improve its classification performance. Finally, we evaluate the performance of our method on real-SAR images, and the experimental results demonstrate that the proposed framework outperforms related methods on classification while reducing the manual annotation substantially.**

***Index Terms*— G^0-based convolutional variational auto-encoder (GCVAE), hybrid network, structural constraint, synthetic aperture radar (SAR) image scene classification.**

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The authors are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China, also with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, Xi'an 710071, China, also with the International Research Center for Intelligent Perception and Computation, Xidian University, Xi'an 710071, China, and also with the Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: f63liu@163.com).

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I. Introduction

The synthetic aperture radar (SAR) image classification is one of the crucial steps in SAR image automatic understanding and interpretation [1]–[3], and many SAR image classification methods have been proposed recently [4]–[6]. We roughly divide them into pixelwise [7], [8] and imagewise [9], [10] classification methods. Pixelwise classification methods generally utilize a huge number of pixelwise labeled samples to train a model or network, yet labeling so many pixelwise samples are time-consuming, laborious, and even requires field investigations. Imagewise methods aim to assign a label to an image based on its geographical attributes and so on. However, due to earth observations, an area of SAR images may contain diverse terrain types, which can form a functional area, such as a residential area or a commercial area, and an SAR image usually includes several different functional scenes. Therefore, imagewise classification results are inadequate for SAR image scene interpretation. Different from the two kinds of classification methods, the goal of scene classification in this article is to assign patchwise labels to various functional areas of an SAR image.

The performance of classical classification methods, such as support vector machines (SVMs) [11], [12], random forests [13], [14], and some other methods [15], is closely related to hand-crafted features, and various feature extraction methods have emerged in recent decades. For example, the gray-level co-occurrence matrix (GLCM) [16] is used to represent the local spatial relations of pixels; the wavelet [17] and Gabor transforms [18] are widely used in the texture analysis of SAR images; and there are also some high-level features, such as bag-of-words [19] and sparse representation [20], [21]. Moreover, various types of features are combined to improve the classification performance [22]. According to extensive domain knowledge and manual ingenuity, these hand-crafted features perform very well on certain applications. However, with the increase of SAR image resolution and scene complexity, these methods become incapable of extracting highly discriminative features for all SAR images, especially for the areas where different terrain types are highly aggregated.

Data-based deep learning methods [23], [24] can automatically extract more abundant and abstract features via convolutional neural networks (CNNs), which have

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achieved state-of-the-art results in many computer vision tasks [25]–[27], including SAR image classification [28]. Zhao *et al.* [29] proposed a discriminant deep belief network for high-resolution SAR image classification, which changed the 2-D images into 1-D vectors, thus losing the spatial information of images. Chen *et al.* [30] utilized the deep convolutional networks in SAR image classification and achieved superior performance over handcrafted features. In general, these methods only employ a large number of labeled samples to train the massive parameters of CNNs in order to avoid overfitting and rarely consider some prior knowledge and related modeling methods of the data.

Model-based classification methods adopt various modeling techniques in light of the data, which have relatively clear physical meaning and strong interpretability. Bayesian estimation [31] is one of the most effective model-based methods, and it can introduce both prior information and hidden variables to calculate the posterior distribution. Unfortunately, the posterior or marginal likelihood is intractable to compute for almost all models. Variational Bayesian inference is a popular approximation solution to tackle these problems [32]. It constructs a variational lower bound for the likelihood through certain assumptions and transforms complex integral problems into optimizing the bound. The most common assumption is that the data or prior follows the Gaussian distribution, and one can derive an analytically tractable solution of the variation lower bound. In our opinion, the choice of prior, which plays a crucial role in the results, should first and foremost consider the characteristics of the data itself, rather than a simple prior for the convenience of calculation and derivation. Moreover, choosing an overly simplistic prior, such as the standard normal distribution, could lead to over-regularization and very poor representations [33], [34].

Frery *et al.* [35], [36] took into account the scattering mechanism and the echo characteristic of SAR images and demonstrated that the probability density function of SAR images obeyed the G^0 distribution, which provides an attractive approach for the statistical modeling of SAR images. Based on this prior knowledge, we employ variational Bayesian inference to derive the analytical expression of the variational lower bound of SAR images [37]. We define this process as G^0-based variational Bayesian inference modeling (GVBI). To realize this model, we design a neural network named the G^0-based convolutional variational auto-encoder (GCVAE), which is a kind of VAE [38], to iteratively optimize the variational lower bound. The GCVAE is a model-based unsupervised generative network that can capture rich distributional information of SAR images. It is noted that the GVBI and GCVAE are further extensions of our previous work [39].

The ultimate objective of this article is to take full advantage of both model-based and data-based methods to improve the classification performance of SAR images using very limited labeled samples. On the one hand, the GCVAE can generate more samples with the same distribution for the CNN to prevent overfitting using very few samples. On the other hand, the data-based CNN can fine-tune the GCVAE under the guidance of labels, so that the distributional characteristics of samples are more conducive to classification. Therefore, we propose a hybrid network that combines the GCVAE and CNN for SAR image scene classification with a few labeled samples, named GCVAE-CNN. The biggest characteristic of the hybrid network is that the generated samples keep the distribution information and diversity under the driving of the GCVAE; meanwhile, the generation process is goal directed of the classification task, rather than generating samples as close to the originals as possible.

We know that deep neural networks mainly capture texture but not structure or shape characteristics. Wu *et al.* [40] and Duan *et al.* [41] hold that the structural information of SAR images is highly important for the SAR scene interpretation, and they proposed the SAR sketch map according to the inherent imaging mechanism, statistical distribution, and geometric characteristics of SAR images, which is a kind of sparse representation of terrain structures of SAR images. Based on the SAR sketching model, we design a sketch fitter (SF) and two structural constraints for GCVAE-CNN, which are used to retain the structural characteristics of SAR images to prevent the performance deterioration of GCVAE caused by CNN's over fine-tuning. Thus, the robustness and accuracy of the hybrid network can be improved further.

The main contributions of this article are as follows.

1) We inherit and expand our previous model [39] and propose a GCVAE for unsupervised learning of the distributional characteristics of SAR images, which is a model-based process.
2) To develop the advantages of both model-based and data-based methods, we construct a hybrid network combining the GCVAE with CNN for SAR image scene classification under very limited labeled samples.
3) Considering that the structural information is very useful for SAR image classification, we design an SF and two structural constraints on both pixel and sketch spaces for the hybrid network to improve its robustness and accuracy.

The remainder of this article is organized as follows. Section II reviews the related work. The complete framework for SAR image scene classification is provided in Section III. The experiments and analysis are presented in Section IV. Finally, the conclusions of this work are presented and future works are discussed in Section V.

II. Related Work

Our approach is closely related to the stochastic gradient variational Bayesian model (SGVB) [38]. The authors derived a differentiable analytical expression of the variational lower bound by introducing the Gaussian prior and a reparameterization trick. Thus, the variational lower bound could be solved by the random gradient descent algorithm. Since the derivation process is a core technology and also a difficult issue of this and related algorithms, we provide some analysis and supplements about the derivation in this section. In addition, the SAR sketching model [40] can sparsely represent the structures of terrain types, which can be used for SAR scene classification; therefore, we also provide a brief introduction about it.

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A. Stochastic Gradient Variational Bayesian Model

Given a data set $X = \{\mathbf{x}^i\}_{i=1}^N$, which consists of N samples, let x and z be the observable and latent variables of the data. Furthermore, let $p_\theta(\mathbf{z})$ and $p_\theta(\mathbf{z}|\mathbf{x})$ be the prior and posterior distributions; $p_\theta(\mathbf{x}|\mathbf{z})$ and $p_\theta(\mathbf{x}, \mathbf{z})$ be the marginal likelihood and joint distributions; and θ be the parameters of the model.

In theory, one can use the Bayesian rule to calculate the posterior distribution, which is then used for optimal decisions, predictions, and so on. Calculating the marginal likelihood or posterior distribution involves very high-dimensional integrals and is analytically intractable [42] in practice. To deal with these problems, instead of directly simplifying the marginal likelihood or posterior distribution, Kingma and Welling [38] employed a more general and efficient stochastic gradient variational Bayesian inference to approximate the log likelihood.

The authors introduced a recognition model $q_\phi(\mathbf{z}|\mathbf{x})$, which is an approximation to the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$, and ϕ be the corresponding model parameters. The log-likelihood can be calculated as follows:

$$\begin{aligned}\log p_\theta(\mathbf{x}) &= \int \log\left(\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})}\right) q_\phi(\mathbf{z}|\mathbf{x})dz \\ &= -D_{\mathrm{KL}}\big(p_\theta(z)||q_\phi(\mathbf{z}|\mathbf{x})\big) + E_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}|\mathbf{z})\big] \\ &\quad + D_{\mathrm{KL}}\big(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})\big). \end{aligned} \tag{1}$$

Since the Kullback–Leibler divergence is nonnegative, $D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) \geq 0$, the variational lower bound of the log-likelihood can be written as

$$L(\mathbf{x};\theta,\phi) = -D_{\mathrm{KL}}\big(p_\theta(\mathbf{z})||q_\phi(\mathbf{z}|\mathbf{x})\big) + E_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x}|\mathbf{z})\big]. \tag{2}$$

To derive the analytical expression of (2), the authors made some assumptions and restrictions. Let the prior is a standard normal distribution, $p_\theta(\mathbf{z}) = N(\mathbf{z}; 0, I)$, the marginal likelihood is a multivariate Gaussian $p_\theta(\mathbf{x}|\mathbf{z}) = N(\mathbf{z}; \mu, \sigma^2 I)$, the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$ follows an approximate Gaussian, and the variational approximate posterior is a multivariate Gaussian distribution $q_\phi(\mathbf{z}|\mathbf{x}) = N(\mathbf{x}; \mu, \sigma^2 I)$. (It should be noted that these assumptions are mainly for the convenience of derivation, not the optimal solution). Then, the first term of (2) is derived as follows [38]:

$$\begin{aligned}&-D_{\mathrm{KL}}\big(p_\theta(\mathbf{z})||q_\phi(\mathbf{z}|\mathbf{x})\big) \\ &\quad = \frac{1}{2}\sum_{j=1}^{J}\Big(1 + \log\Big(\big(\sigma_j\big)^2\Big) - \big(\mu_j\big)^2 - \big(\sigma_j\big)^2\Big) \end{aligned} \tag{3}$$

where J represents the dimension of hidden variables z.

For the second term of (2), the authors utilized a reparameterization trick, $\mathbf{z} = \mu + \sigma \odot \varepsilon$, where $\varepsilon = N(0, 1)$, and $\odot$ denotes the elementwise product. The expectation integral can be calculated using a Monte Carlo approximation [36]

$$E_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(x|z)\big] \approx \frac{1}{N}\sum_{l=1}^{N}\big(\log p_\theta\big(x|z^{(l)}\big)\big) \tag{4}$$

where N denotes the number of observable variables.

According to (3) and (4), they obtained the whole analytical expression of the variational lower bound, which is differentiable. Therefore, all the parameters of the model could be optimized jointly by a neural network with the random gradient descent algorithm. The variational lower bound was used as the loss function of the neural network. Specifically, (3) was regarded as the regularization term; for $p_\theta(\mathbf{x}|\mathbf{z})$ was assumed to be a Gaussian distribution, where the mean was a function $\mu(\mathbf{z})$ and the variance was a constant σ_0^2, (4) was approximated as a mean-square term as follows:

$$\log p_\theta(\mathbf{x}|\mathbf{z}) \approx \frac{1}{2\sigma_0^2}\|\mathbf{x} - \mu(\mathbf{z})\|^2. \tag{5}$$

Finally, the authors constructed a VAE to optimize the variational lower bound iteratively.

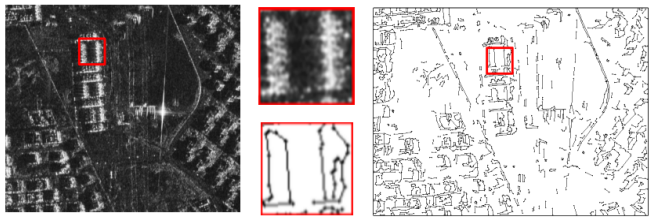


Fig. 1. SAR image and sketch map. On the left-hand side is the SAR image, on the right-hand side is the sketch map, and in the middle are two enlarged image patches.

B. SAR Sketching Model

On the basis of Marr's vision [43] and the initial sketching model for optical images that was proposed by Guo *et al.* [44], Liu *et al.* [41] investigated the edge and line detection methods of SAR images and proposed the SAR sketching model and extracting method of the SAR sketch map. Sketch line segments with direction are the primitives of the sketch map, as shown in Fig. 1. Based on the SAR sketching model, Liu *et al.* [45] proposed a hybrid method of SAR speckle reduction that was based on geometric-structural blocks and adaptive neighborhoods. Liu *et al.* [46] endowed the sketch line segments with semantic information and proposed a hierarchical visual semantic model, which was used for polarization SAR image classification. These semantic sketch line segments can sparsely represent various land-cover types and reflect the structural characteristics of SAR images. For example, some sketch line segments indicate objects that are high above the ground, such as the edges that are generated by the bright spots of a building and its shadows; some represent the boundaries of two different objects; and some denote line targets, such as bridges and roads. Thus, we can use the SAR sketching model to constrain the network to capture the structural characteristics of SAR images.

III. Proposed Method

A. Overview of the Proposed Method

In this section, we introduce the whole network framework of the proposed method. To take full advantage of both model-based and data-based methods, we construct a hybrid network with GCVAE and a common CNN for SAR image scene classification. On the one hand, the GCVAE can provide diverse samples for the CNN to relieve overfitting due to a few

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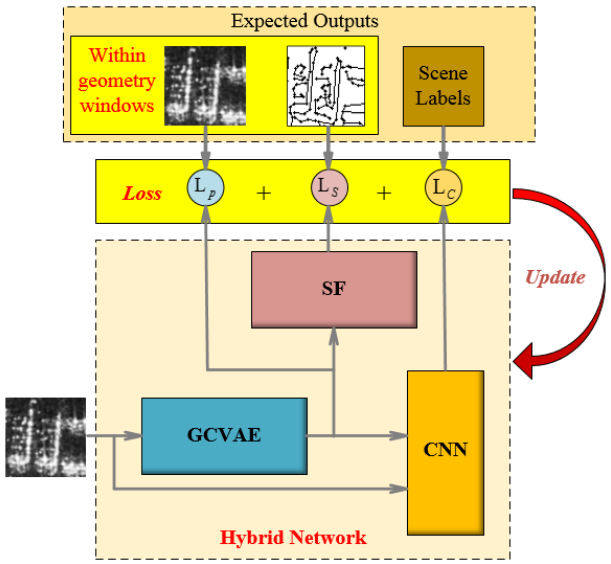


Fig. 2. Overall flowchart of the hybrid network with structural constraints.

labeled samples. On the other hand, the CNN can adjust the GCVAE adaptively with the guidance of labels to generate highly discriminative samples that are more conducive to classification. The idea of joint optimization of the GCVAE and CNN is also inspired by semi-GAN [47] and Auxiliary Classifier Generative Adversarial Networks (ACGAN) [48], where all three components (generator, discriminator, and classifier) iteratively make each other better. In addition, two structural constraints are introduced into the hybrid network for capturing the structural information, which is very helpful for SAR image classification. The overall flowchart of the hybrid network with structural constraints is shown in Fig. 2.

B. GCVAE

Model-based approaches usually have a solid theoretical foundation. In this section, we first introduce the mathematical derivation of GCVAE and then give its concrete implementation.

Frery *et al.* [35] assumed that the background of SAR images obeyed the generalized inverse Gaussian and that the coherent speckle followed the Gamma distribution. They demonstrated that the general probability statistical model of SAR images followed the G^0 distribution. The probability density function of G^0 distribution is defined as

$$P^{G^0}(I(x,y);\alpha,\gamma,n)=\frac{n^n\Gamma(n-\alpha)I(x,y)^{n-1}}{\gamma^\alpha\Gamma(n)\Gamma(-\alpha)(\gamma+nI(x,y))^{n-\alpha}}\quad \text{s.t. } -\alpha,\gamma,I(x,y)>0 \tag{6}$$

where $I(x,y)$ denotes the gray value of pixels. n represents the number of looks. To restrain the influence of speckle noises, an SAR image is generated by averaging n statistically independent images (looks). When $n=1$, the speckle noises of SAR images are the largest. γ is the scale parameter that reflects the average energy of SAR images. α is the uniformity that is related to the uniformity of terrain types, and $\Gamma(x)$ denotes the Gamma law [35]. The G^0 distribution can be expressed as the beta-prime distribution under the condition of $n=1$, which is written as follows:

$$P^{\beta'}(I(x,y);\alpha,\gamma,n)=\frac{-\alpha\gamma^{-\alpha}}{(\gamma+I(x,y))^{1-\alpha}}\quad \text{s.t. } -\alpha,\gamma,I(x,y)>0. \tag{7}$$

Based on the prior information of distribution and our previous work [39], we exhibit and extend the derivation process of the analytical expression of the variational lower bound in this article. Owing to just the first term of the variational lower bound (2) is related to the distribution of data, we derive this term in detail, which can be expressed as

$$\begin{aligned}&-D_{KL}\big(p_\theta(\mathbf{z})||q_\phi(\mathbf{z}|\mathbf{x})\big)\\&\quad=\int q_\theta(\mathbf{z})\log p_\theta(\mathbf{z})d\mathbf{z}-\int q_\theta(\mathbf{z})\log q_\theta(\mathbf{z})d\mathbf{z}.\end{aligned} \tag{8}$$

Let prior follows the G^0 distribution, $p_\theta(\mathbf{z})=G^0(\mathbf{z};c,\gamma_1)$, where c and γ_1 denote the uniformity and scale parameters, respectively; and $q_\phi(\mathbf{z}|\mathbf{x})=G^0(\mathbf{x};\alpha,\gamma)$. We derive the two parts of (8) in two cases ($n=1$, and $n\neq 1$). When the number of looks $n=1$, the derivation of $\int q_\theta(\mathbf{z})\log p_\theta(\mathbf{z})d\mathbf{z}$ is as follows:

$$\begin{aligned}&\int q_\theta(\mathbf{z})\log q_\theta(\mathbf{z})d\mathbf{z}\\&=\int P^{G^0}(\mathbf{z};-\alpha,\gamma)\log P^{G^0}(\mathbf{z};-\alpha,\gamma)d\mathbf{z}\\&=\log(-\alpha)-\alpha\log\gamma+\frac{(1-\alpha)\log(b+\gamma)(b+\gamma)^\alpha}{\gamma^\alpha}\\&\quad+\frac{(1-\alpha)(a+\gamma_1)^\alpha}{\alpha\gamma^\alpha}-\frac{(1-\alpha)(b+\gamma)^\alpha}{\alpha\gamma^\alpha}\\&\quad-\frac{(1-\alpha)\log(a+\gamma)(a+\gamma)^\alpha}{\gamma^\alpha}.\end{aligned} \tag{9}$$

When $n=1$, the derivation process of $\int q_\theta(\mathbf{z})\log q_\theta(\mathbf{z})d\mathbf{z}$ is as follows:

$$\begin{aligned}&\int q_\theta(\mathbf{z})\log q_\theta(\mathbf{z})d\mathbf{z}\\&=\int P^{G^0}(\mathbf{z};-\alpha,\gamma)\log P^{G^0}(\mathbf{z};-\alpha,\gamma)d\mathbf{z}\\&=\log(-\alpha)-\alpha\log\gamma+\frac{(1-\alpha)\log(b+\gamma)(b+\gamma)^\alpha}{\gamma^\alpha}\\&\quad-\frac{(1-\alpha)\log(a+\gamma)(a+\gamma)^\alpha}{\gamma^\alpha}+\frac{(1-\alpha)(a+\gamma_1)^\alpha}{\alpha\gamma^\alpha}\\&\quad-\frac{(1-\alpha)(b+\gamma)^\alpha}{\alpha\gamma^\alpha}.\end{aligned} \tag{10}$$

Equations (4), (9), (10) are brought into (2) to obtain the complete analytical expression of the GVBI in the case $n=1$, as follows:

$$\begin{aligned}L(\mathbf{x};\theta,\phi)=\sum_{j=1}^{J}\big(&\log c+c\log\gamma_1-\log(-\alpha_j)+\alpha_j\log\gamma_j\\&+\frac{(1+c)\log(b+\gamma_1)(b+\gamma_1)^{\alpha_j}}{{\gamma_j}^{\alpha_j}}\\&-\frac{(1+c)\log(a+\gamma_1)(a+\gamma_1)^\alpha}{\gamma^\alpha}\end{aligned}$$

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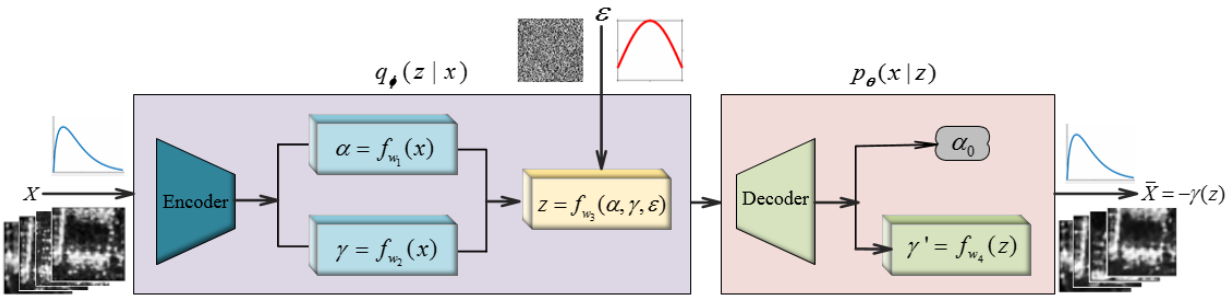


Fig. 3. Overview of the proposed GCVAE, where f is fitted by fully connected layer network, and (w_1, w_2, w_3, w_4) are their parameters.

$$
\begin{aligned}
&+\frac{(1+c)(a+\gamma_1)^{\alpha_j}}{\alpha_j {\gamma_j}^{\alpha_j}}-\frac{(1+c)(b+\gamma_1)^{\alpha_j}}{\alpha_j {\gamma_j}^{\alpha_j}}\\
&-\frac{(1-\alpha_j)\log(b+\gamma_j)(b+\gamma_j)^{\alpha_j}}{{\gamma_j}^{\alpha_j}}\\
&+\frac{(1-\alpha_j)\log(a+\gamma_j)(a+\gamma_j)^{\alpha_j}}{{\gamma_j}^{\alpha_j}}\\
&-\frac{(1-\alpha_j)(a+\gamma_j)^{\alpha_j}}{\alpha_j {\gamma_j}^{\alpha_j}}\\
&+\frac{(1-\alpha_j)(b+\gamma_j)^{\alpha_j}}{\alpha_j {\gamma_j}^{\alpha_j}}+\frac{1}{N}\sum_{l=1}^{N}\left(\log p_\theta\left(x|z^{(l)}\right)\right)\\
&\text{s.t. } n=1,\ 0<a<b\le 1 \qquad (11)
\end{aligned}
$$

where a and b are the upper and lower bounds of the normalized gray value of pixels, N is the number of observable variables, and J denotes the dimension of hidden variables. When the number of looks $n \neq 1$, the derivation process of (8) is similar, and we do not show it here anymore.

After deriving the differentiable analytical expression of the variational lower bound, it can be used as the loss function of a neural network. In our previous work, the first several terms of the variational lower bound were used as regularizations, and the last term was considered as a mean square error. For the probability density function of G^0 is different from the Gaussian distribution, we update the last term in this work. It should be expressed as an absolute loss between x and $-\gamma(\mathbf{z})$.

Let $p_\theta(\mathbf{x}|\mathbf{z})$ be a G^0 distribution $p_\theta(\mathbf{x}|\mathbf{z}) = G^0(\mathbf{z}; \alpha, \gamma)$, when the number of looks $n = 1$, its probability density function is as in (7); assume the parameter of uniformity is a constant α_0, and the last term of variational lower bound can be approximated as follows:

$$
\begin{aligned}
\log p_\theta(x|z) &= \log \frac{-\alpha_0 * \gamma(z)^{-\alpha_0}}{(x+\gamma(z))^{1-\alpha_0}}\\
&\approx (\alpha_0 - 1)\log\|\mathbf{x} - (-\gamma(\mathbf{z}))\|_1. \qquad (12)
\end{aligned}
$$

After the theoretical derivation, we introduce the concrete realization of the model. According to the GVBI, we construct a GCVAE, and the overview of the network structure is shown in Fig. 3. It mainly consists of two parts: recognition module $q_\phi(\mathbf{z}|\mathbf{x})$ and generation module $p_\theta(\mathbf{x}|\mathbf{z})$.

The recognition module recognizes the two parameters α and γ from the input samples x using a five-layer convolutional network as the encoder (three convolutional layers and two fully connected layers). The two parameters, together with the auxiliary variable ε, are used to calculate the hidden variables z. The generation module generates γ' from the hidden variables using a five-layer transposed convolutional network as a decoder, where parameter α^0 is restricted to a constant. All the parameters $(\alpha, \gamma, \phi, \theta, w)$ of GCVAE are learned jointly by the loss function (11). We use the original image patches as the input of GCVAE, and the output is $-\gamma(\mathbf{z})$.

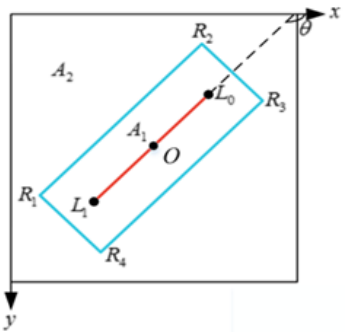


Fig. 4. Illustration of the geometry window.

C. Two Structural Constraints

Structural information of SAR images is crucial for SAR image classification. Based on the sketch map, which can sparsely represent the structural information of SAR images, we design two structural constraints to assist the hybrid network to capture the structural information from both pixel and sketch spaces.

1) Structural Constraint on Pixel Space: A sketch map consists of many sketch line segments with orientation, which can represent the location of the terrains and their shadows. Therefore, we design a geometry window along each sketch line segment and construct a structural constraint to guarantee the consistency of structures between SAR and their generative image patches within the window.

The geometry window is shown in Fig. 4. Let S be a sketch map with one sketch line segment L_0L_1, where O and θ represent the midpoint and orientation, respectively. We build a rectangle $R_1R_2R_3R_4$ as its geometry window, O is the center of the rectangle, the orientation of R_1R_2 is the same as L_0L_1, and the distance L_0 to R_1R_2 is equal to the distance L_0 to

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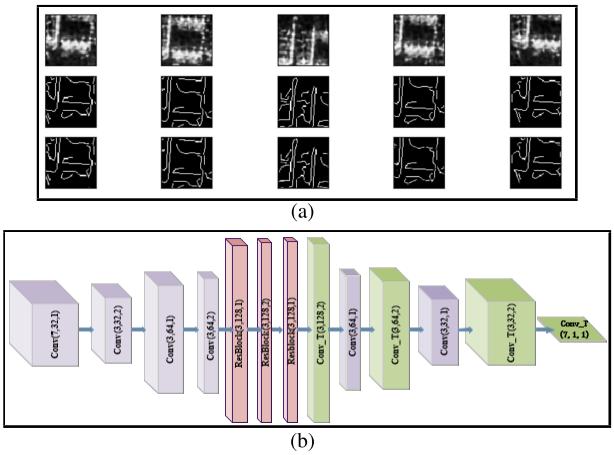


Fig. 5. I/O and network structure of SF. (a) Input, expected output, and actual output of SF, where the first line is five input image patches, the second line is the expected output, and the third line is the actual output. (b) Network structure of SF, where Conv (7, 32, 1) represents convolutional layer parameters (kernel size, filter number, and strides).

R_2R_3 (the distance is set as three pixels). Let the value of area A_1 be 1 (within the window), and the value of area A_2 be 0, which form a mask matrix of the sketch map. When there are L line segments in a sketch map S, we can construct L geometry windows and L mask matrices $\{M_1, M_2, \ldots, M_L\}$. Based on the mask matrices $L_P(\mathbf{x}; \varpi_p)$, we design a structural constraint on pixel space, which is defined as follows:

$$L_P(\mathbf{x}; \varpi_p) = \arg\min_{\varpi_p} \sum_{i=1}^{L} \sum_{j=1}^{N} \left\| \left(\mathbf{x}^j - \left(G\left(\mathbf{x}^j\right)\right)\right) * M_i \right\|_2. \tag{13}$$

2) Structural Constraint on Sketch Space: We name the space represented by sketch line segments as the sketch space. To capture more structural information, we also constrain sketch line segments on the sketch space. To achieve this aim, we must obtain the sketch map of original and generated SAR image patches. The former can be calculated by the SAR sketching model [40]. For the latter, we design a neural network, named SF, to fit a sketch extraction function $p_s(i, j) = S(X(i, j))$, where X represents the SAR image patch and $p_s(i, j)$ denotes the sketchable probability of each pixel in the patch. With the loss of binary cross-entropy, we can train the SF to approach the sketch extraction function infinitely. The network structure and I/O of SF are shown in Fig. 5. The SF consists of six convolutional layers, three residual blocks, and four transposed convolutional layers. We use the original image patches as the input, the accordingly sketch map patches as the expected output of SF, and the actual output is the sketchable probability, as shown in Fig. 5(a).

After calculating the sketchable probability, we can obtain a binary image as the generated sketch map. Finally, we design a structural constraint $L_S(\mathbf{x}, s; \varpi_s)$ on the sketch space, which is written as follows:

$$L_S(\mathbf{x}, s; \varpi_s) = \arg\min_{\varpi_s} \sum_{i=1}^{L} \sum_{j=1}^{N} \left\| \left(s_j - \left(SF\left(G\left(\mathbf{x}_j\right)\right)\right) \odot M_i \right\|_1 \tag{14}$$

where ϖ_s denote related parameters, s_j is the original sketch map, $\mathbf{x}_j$ is the original SAR image patch, $G(\mathbf{x}_j)$ denotes the generated sample by the GCVAE, and $SF(G(\mathbf{x}_j))$ represents the generated sketch map of $G(\mathbf{x}_j)$ by the SF.

D. Summary of the Algorithm

We summarize the whole algorithm in this section. The training of the proposed network framework includes three steps. First, solely training GCVAE until it converges. Second, training SF alone until it converges. Finally, unified training the hybrid network, including GCVAE, SF, and CNN. The detailed algorithm description is listed in Algorithm 1. After training, we can use the hybrid network to predict the classification results directly. The total loss function of the hybrid network consists of three parts: classification loss (cross entropy) and two structural constraint losses, which is written as follows:

$$L_{\text{all}} = L_c(\mathbf{x}, y; \omega) + \beta_1 \cdot L_p\left(\mathbf{x}; \varpi_p\right) + \beta_2 \cdot L_s(\mathbf{x}, s; \varpi_s) \tag{15}$$

where β_1 and β_2 are two hyperparameters to balance the proportions of the two structural constraint losses.

Compared with pixelwise methods, our patchwise classification method has better regional consistency and higher efficiency, but the accuracy of edge, particularly the localization of line targets, is not accurate enough. To more accurately locate line targets and further optimize the regional consistency, we inherit some of the existing work of our team [49]. Under the guidance of sketch map [40] and regional map [50], we locate some line targets and set different parameters for the aggregated regions. We also provide the final results in the experiments.

IV. Experiments and Analysis

A. Experimental Data and Settings

We validate the performance of the proposed method on four real SAR images, and their detailed information is listed in Table I. To reduce the manual annotation as much as possible, we label one large patch for each scene of an image as the training areas, where the patch size of the first two images is 96×96 and others are 192×192. The selected labeled areas are marked with colored square or rectangle boxes on the original SAR images, as shown in Figs. 7(a) and 10(a)–12(a). In addition, we introduce a data augmentation scheme that scale and rotate the large patches, including magnifying $2\times$, shrinking $2\times$ and rotating seven different directions $\{45°, 90°, 135°, 180°, 225°, 270°, 315°\}$. Then, we randomly sample 256 small patches of size 24×24 on each augmented patch, which is used as the training samples. Moreover, in order to quantitatively evaluate the proposed approach, we select several typical scenes of size 192×192 from different SAR images and construct two data sets, as shown in Figs. 6 and 8. Then, 2000 patches of size 24×24 are randomly sampled for each scene, where 256 patches are used as the training samples and the rest are the testing samples. We define the classification accuracy (CA) of a class as the ratio of the number of patches correctly classified for the class to the test number of patches in this class. The overall

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TABLE I
DETAILED INFORMATION OF SAR IMAGES

Image	Size	Resolution	Band	Modality	Location	Sensor
Piperiver	600×432	$1\ m$	*Ku*-band	Stripmap	New Mexico	Airborne SAR
Bridge	350×768	$1\ m$	*X*-band	Spotlight	New Mexico	Airborne SAR
Noerdlin	1308×1452	$1\ m$	*X*-band	Spotlight	Swabian Jura	TerraSAR
W. DC	2000×810	$1\ m$	*Ku*-band	Stripmap	Washington D.C.	Airborne SAR

CA (OA) is defined as the ratio of the number of correctly classified patches in the whole class to the total test number of patches, and the average CA (AA) is the average of the CA of each class. We also calculated the Kappa coefficient (Kappa) of each method. Given that the design of the CNN is not the focus of this article, and the labeled samples are very limited, we choose the LeNet-5 for all involved CNNs. The hyperparameters β_1 and β_2 are set as 0.7 and 0.5, respectively, and the number of looks is set as $n = 1$. The initial learning rate is set as 0.001, the GCVAE, SF, and hybrid network are trained for 50, 100, and 50 epochs, respectively, GAN is trained for 500 epochs, and other settings for training the networks are similar to those in [51]. All the experiments are implemented on a personal computer with a single Intel Core i5-4590 CPU and an NVIDIA Quadro k620 GPU. The implementation environment is Python 3.5.

Algorithm 1 The Training of the Proposed Method

1: Training GCVAE:
  1.1 Input: The original SAR image patches;
  1.2 Expected output: The original SAR image patches;
  1.3 Actual output: The generated SAR image patches;
  1.4 Iterative optimizing the GCVAE until the objective function (11) is minimized.
2: Training SF:
  2.1 Input: Tthe original SAR image patches;
  2.2 Expected output: The sketch patches of corresponding SAR image patches;
  2.3 Actual output: The generated sketch patches, that is sketchable probability of original patches;
  2.4 Iterative optimizing the SF with binary cross-entropy.
3: Training hybrid network:
  3.1 Input: The labeled patches of an SAR image;
  3.2 Expected output: The original patches within geometry windows, sketch patches within geometry windows, and corresponding labels;
  3.3 Actual output: The generated patches, sketchable probability of original patches, and category probability of patches;
  3.4 Iterative optimizing the hybrid network until the loss function (15) is minimized.

To demonstrate the effectiveness of the two main components of the hybrid network, the GCVAE and structural constraints, we first perform an ablation study in Section IV-B. Then, parameter analysis is conducted in Section IV-C. Finally, in Sections IV-D and IV-E, we quantitatively and qualitatively test the proposed approach by comparison with seven related methods listed as follows.

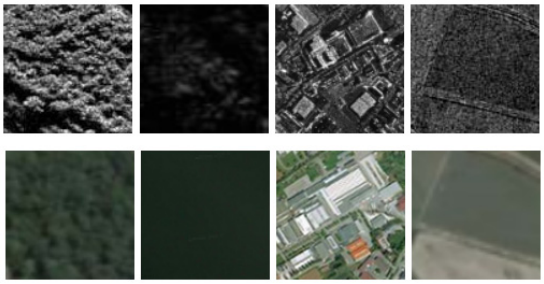


Fig. 6. Four scenes data set (forest, water, industrial, and grassland). The first line shows the original patches. The second line is the optical image patches.

1) GLCM-SVM: We extract 16-dimension GLCM features for each small patch and employ an SVM as the classifier.
2) Gabor-SVM: This method applies Gabor transform to extract features, and each small patch is represented by 24-dimension Gabor features.
3) CNN: This method applies LeNet-5 for SAR image scene classification.
4) Gauss-CNN: The Gaussian hybrid network that combines the Gaussian-based VAE with the CNN for SAR image scene classification.
5) Gauss-CNN-Two: The Gauss-CNN with two structural constraints.
6) GCVAE-CNN: Our hybrid network without structural constraints.
7) GAN-CNN-Two: Replace GCVAE with DCGAN in the hybrid network with two structural constraints.

B. Ablation Study

In this section, two groups of data are utilized in the ablation study. We use the data of four scenes [Fig. 6] to calculate the numerical results, including CA of each scene, AA, OA, and Kappa. The bridge image [Fig. 7(a)] is used to qualitatively illustrate the influence of the two key components for the hybrid network. In addition, we conduct five experiments, including CNN, Gauss-CNN, GCVAE-CNN, GCVAE-CNN-One (using our hybrid network with the structural constraint on pixel space), and GCVAE-CNN-two to evaluate the effectiveness of the GCVAE and structural constraints. The results are shown in Table II and Fig. 7.

Table II lists the numerical classification results for the five compared experiments. The CA of all scenes in the CNN

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TABLE II
ABLATION STUDY OF THE PROPOSED METHOD (%)

Methods	Forest	Water	Industrial	Grassland	AA	OA	Kappa
CNN	86.55	94.25	84.01	88.82	88.41	88.41	0.8326
Gauss-CNN	88.69	94.69	88.92	90.97	90.82	90.82	0.8467
GCVAE-CNN	90.58	95.16	91.67	92.89	92.58	92.58	0.8836
GCVAE-CNN-One	92.52	95.19	93.06	93.94	93.68	93.68	0.9024
GCVAE-CNN-Two	**93.61**	**95.22**	**94.53**	**94.67**	**94.51**	**94.51**	**0.9130**

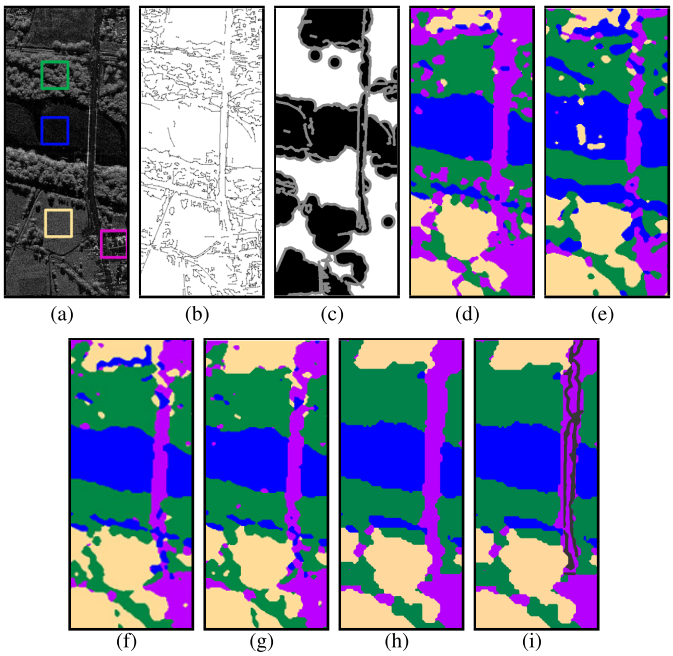


Fig. 7. Ablation study of the proposed method. (a) Bridge image that contains four semantic scenes: residential (magenta), forest (green), water (blue), and land (light yellow). 350 × 768, X-band, and 1-m resolution. (b) Sketch map (25, 1.5, and 0.7). (c) Regional map ($K = 7$). (d) CNN. (e) Gauss-CNN. (f) GCVAE-CNN. (g) GCVAE-CNN-One. (h) GCVAE-CNN-two. (i) Final result.

method is the lowest. This phenomenon is mainly due to two reasons: one is that different scenes have some similarity in both the texture and structure characteristics. For example, it is very difficult to distinguish the forest and industrial scenes with the CNN only. In addition, the number of labeled samples is very limited, which may be insufficient to train an excellent classification network. To solve these problems, the model-based method and structural constraints are employed in the CNN. The result of the Gauss-CNN is better than that of the CNN, which indicates that the combination of model-based and data-based methods is effective for the scene classification task. It is noticeable that the AA and OA of GCVAE-CNN are 1.76% higher than Gauss-CNN, the Kappa coefficient is 0.0369 higher than Gauss-CNN. This is because we consider the prior information of SAR in the process of variational Bayesian inference modeling. Compared with the Gauss-based VAE, GCVAE is more suitable for SAR scene classification. We can see from Table II that the AA and OA of the proposed method outperform the CNN, Gauss-CNN, GCVAE-CNN, and GCVAE-CNN-one by 6.1%, 3.69%, 1.93%, and 0.83%, respectively, the result of Kappa is also higher than other methods. The results validate that the hybrid network with the two structural constraints is quite effective, especially for the forest and industrial scenes.

Fig. 7 shows the qualitative results of the ablation study. Fig. 7(a) is the original SAR image, where the four colored square boxes are the training areas that we manually labeled. Fig. 7(b) shows the sketch map of the bridge image. Fig. 7(c) presents the regional map. Fig. 7(d) displays the classification result of the CNN, where numerous misclassified patches can be observed, particularly for the residential scene. Fig. 7(e) shows the classification result of the Gauss-CNN, we can find that many shadows of forest and land scenes are misclassified as waters. The classification result using our hybrid network without structural constraints is shown in Fig. 7(f); we can discover that the misclassifications of residential and water scenes are mitigated when compared with the Gauss-CNN, which shows the stronger feature representation of GCVAE. The classification result of the GCVAE-CNN with the structural constraint on pixel space is presented in Fig. 7(g), we can see that the classification result is better than that in Fig. 7(f), and the misclassifications of waters are further reduced. Fig. 7(h) shows the result of the proposed method, notably, fewer isolated patches and better spatial consistency can be observed in Fig. 7(h), and our method exhibits better performance. The results of Fig. 7(h) and (g) further justify that the structural constraints can effectively improve the classification performance of the hybrid network. Fig. 7(l) is the final classification result where the boundary of the line target (road) is located under the guidance of a sketch map and regional map.

C. Parameter Analysis

In this section, we analyze four main parameters: the size of image patches, the number of labeled samples, and hyperparameters β_1 and β_2. In theory, the larger the patch, the more information it contains. The more samples, the better the classification performance of the network has. However, the size of manually labeled large patches is fixed in our experiments; therefore, the two parameters cannot be increased at the same time. The two hyperparameters β_1 and β_2 determine the proportions of different structural constraints in the total loss function of the hybrid network, which will affect the classification performance to some extent. In addition, we calculate the floating point of operations (FLOPs), number of trainable parameters and inference time of an image for four

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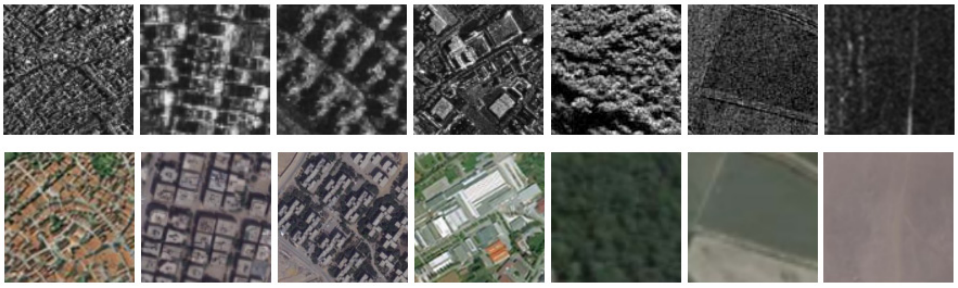


Fig. 8. Seven scenes data set (dense residential, medium residential, sparse residential, industrial, forest, grassland, and idle-land). The first line presents the original patches. The second line presents their corresponding optical image patches.

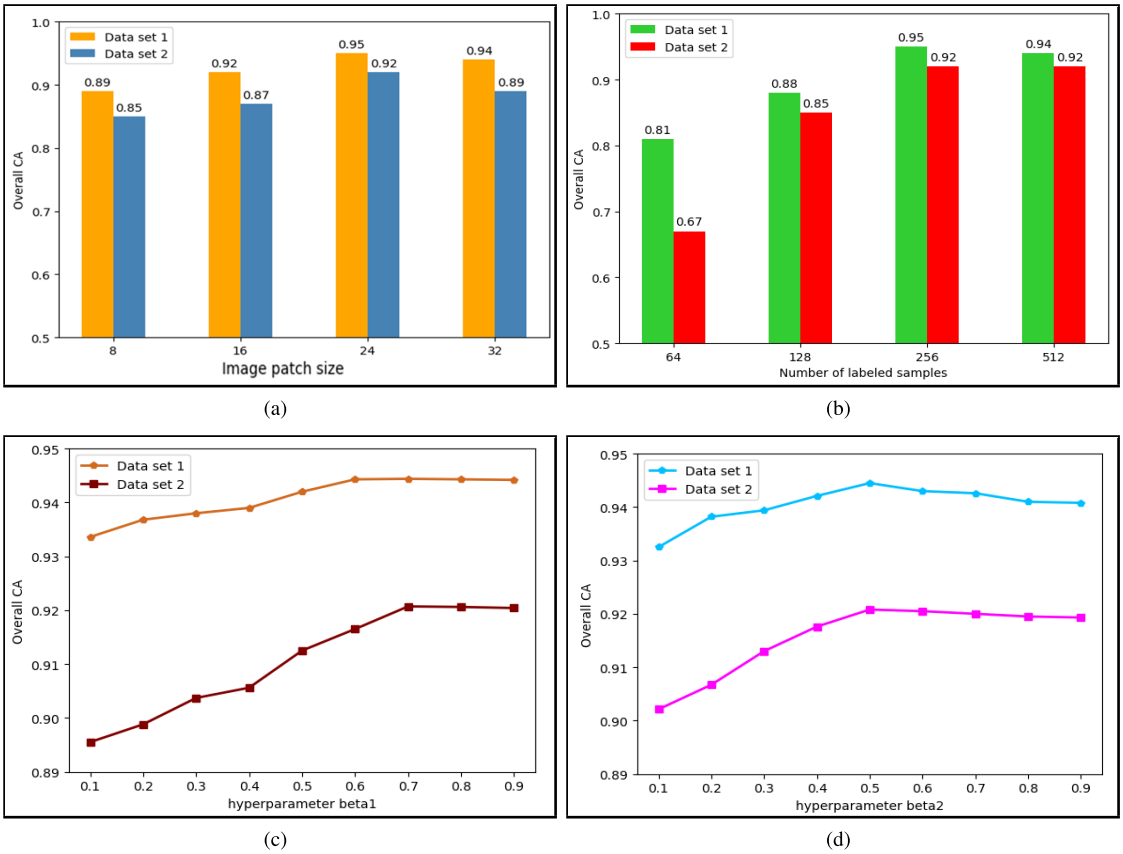


Fig. 9. Influence of different parameters on OAs. (a) Size of image patches. (b) Number of labeled samples. (c) Hyperparameter β_1. (d) Hyperparameter β_2.

TABLE III
PARAMETERS, FLOPS, AND INFERENCE TIME OF FOUR NETWORKS USED IN THE EXPERIMENTS

Methods	Params (M)	FLOPs (M)	Time (ms)
CNN	5.23	10.45	774
Gauss-CNN-Two	9.42	18.84	1908
GAN-CNN-Two	9.83	19.65	1627
GCVAE-CNN-Two	9.42	18.84	1908

networks involved in the experiments, and the experimental results are shown in Table III.

We use two sets of data [Figs. 6 and 8] to calculate the OA values under various parameter values. Data set of four and seven scenes are recorded as data set 1 and data set 2, respectively. In addition, the value range of the four parameters are determined according to the size of SAR images and the network scale, where the size of image patches is set as $\{8, 16, 24, 32\}$, the number of labeled samples is set as $\{64, 128, 256, 512\}$, and the hyperparameters β_1 and β_2 take values from 0 to 1 with a step size of 0.1. Each OA is the average of ten runs, and the experimental results are plotted in Fig. 9.

Fig. 9 shows the influence of the four parameters on the OA values. It can be observed obviously from Fig. 9(a) that when the size of image patches is 8, the OA is the lowest, and the OA is the highest when the size of image patches is 24 on the two data sets. Therefore, the size of image patches is set to 24 in our approach. Fig. 9(b) shows the influence of the number of labeled samples on the OA values. When the number of labeled samples is less than 256, the OA significantly increases with the increment of the number. When it is more than 256, the OA slightly decreases. Therefore, the number of labeled samples is set as 256 in the following

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TABLE IV
CA(%), AA(%), OA(%), AND KAPPA OF FOUR SCENES DATA SET WITH DIFFERENT METHODS

Methods	Forest	Water	Industrial	Grassland	AA	OA	Kappa
GLCM-SVM	80.25	88.79	84.35	84.75	84.54	84.54	0.7989
Gabor-SVM	83.79	**95.58**	82.09	85.68	86.79	86.79	0.8152
CNN	86.55	94.25	84.00	88.82	88.41	88.41	0.8326
Gauss-CNN	88.69	94.69	88.92	90.97	90.82	90.82	0.8467
Gauss-CNN-Two	91.62	95.07	91.58	93.65	92.98	92.98	0.8891
GCVAE-CNN	90.58	95.16	91.67	92.89	92.58	92.58	0.8836
GAN-CNN-Two	92.47	95.41	93.55	93.74	93.79	93.79	0.9025
GCVAE-CNN-Two	**93.61**	95.22	**94.53**	**94.67**	**94.51**	**94.51**	**0.9130**

TABLE V
CA(%), AA(%), OA(%), AND KAPPA OF SEVEN SCENES DATA SET WITH DIFFERENT METHODS

Methods	D.R	M.R	S.R	Industrial	Forest	Grassland	Idle-land	AA	OA	Kappa
GLCM-SVM	72.22	74.78	76.55	77.50	80.75	77.35	84.09	77.61	77.61	0.7321
Gabor-SVM	76.53	80.06	78.32	82.11	78.32	80.76	80.34	79.49	79.49	0.7562
CNN	79.19	83.52	79.61	84.08	81.68	82.58	85.56	82.32	82.32	0.7859
Gauss-CNN	82.35	84.21	83.57	84.26	84.09	85.44	90.39	84.90	84.90	0.8026
Gauss-CNN-Two	87.11	89.66	84.09	86.74	89.54	89.16	91.96	88.32	88.32	0.8335
GCVAE-CNN	88.56	90.01	87.25	91.52	90.01	91.31	**92.48**	90.16	90.16	0.8592
GAN-CNN-Two	89.34	**90.78**	89.06	91.67	91.79	92.57	92.31	91.07	91.07	0.8853
GCVAE-CNN-Two	**90.37**	90.52	**90.48**	**92.36**	**92.37**	**92.78**	92.39	**91.61**	**91.61**	**0.8926**

experiments. The influence of β_1 on OA is shown in Fig. 9(c). It can be observed that the OA increases significantly with the increment of and roughly converges at 0.7. When $\beta_1 > 0.7$, the OA will no longer grow. Thus, β_1 is set to 0.7 in our method. Fig. 9(d) shows the curve of OA with various parameters β_2. The OA increases as the parameter values are increased. When $\beta_2 = 0.5$, the accuracy rises to the highest value and subsequently declines. Thus, β_2 is set to 0.5 according to the experimental result.

From Table III, we can see that the FLOPs, parameters, and run time of the three hybrid networks are higher than the single-CNN classifier. For the network structure of Gauss-CNN-two and our method is the same, they have the same indexes. The structure of the generator and discriminator in GAN-CNN-two are different from VAE, and we use only the generator during the test phase; therefore, its indexes of FLOPs and number of parameters are higher, and the run time is slightly lower compared with the proposed method.

D. Quantitative Analysis of SAR Images

We use two data sets and seven compared experiments to quantitatively analyze the classification performance of the proposed method. We calculate the CA of a class, AA, OA, and Kappa ten times for each method, and the average results are shown in Tables IV and V.

The experimental results of all the methods on four scenes data set are listed in Table IV. We can see that the CA values of the Gabor-SVM are better than those of the GLCM-SVM, especially for the accuracy of the water scene, which is the highest of the eight methods. This result occurs because the Gabor transform takes the characteristics of direction and scale into account and extracts more abundant features. The AA and OA of the CNN obtain a slightly better performance than the results of the Gabor-SVM. The classification results of the Gauss-CNN are superior to the results of the CNN, which confirms that a hybrid network is an effective approach for scene classification under limited labeled samples. The CA of each class and the AA, OA, and Kappa of the Gauss-CNN-Two are higher than the results of the Gauss-CNN. These results show that the proposed structural constraints are efficient to improve the classification performance of the hybrid network. The AA and OA of the GCVAE-CNN are 1.76% higher than that of the Gauss-CNN, which indicates that the samples generated by the GCVAE are more discriminative for the SAR image classification. Compared with the other methods, the proposed method acquires the highest AA, OA, and Kappa.

In Table V, D.R, M.R, and S.R represent dense, medium, and sparse residentials, respectively. According to the results, it can be seen that the Gabor-SVM obtains slightly better results than the GLCM-SVM. The AA, OA, and kappa of the CNN are higher than that of the Gabor-SVM and GLCM-SVM, especially the medium residential and industrial scenes, which indicates that the neural network can capture more discriminative features and obtains better classification performance than the hand-crafted features. However, the AA, OA, and Kappa of the CNN are lower than the results of the Gauss-CNN. This is because the training samples of these seven classes are too limited for CNN to adequately train the deep network. Compared with the Gauss-CNN, the Gauss-CNN-Two employs two structural constraints to optimize the hybrid network, and the classification performance is greatly improved, the AA and OA increased by 3.42%, especially the dense and medium residential scenes by 4.76% and 5.45%, respectively. In addition, we can see that the AA, OA, and Kappa of the GCVAE-CNN are higher than the results of the Gauss-CNN and Gauss-CNN-two. The CA of the idle land scene in GCVAE-CNN is the highest, and the CA of medium residential in GAN-CNN-two is the highest. The AA and OA

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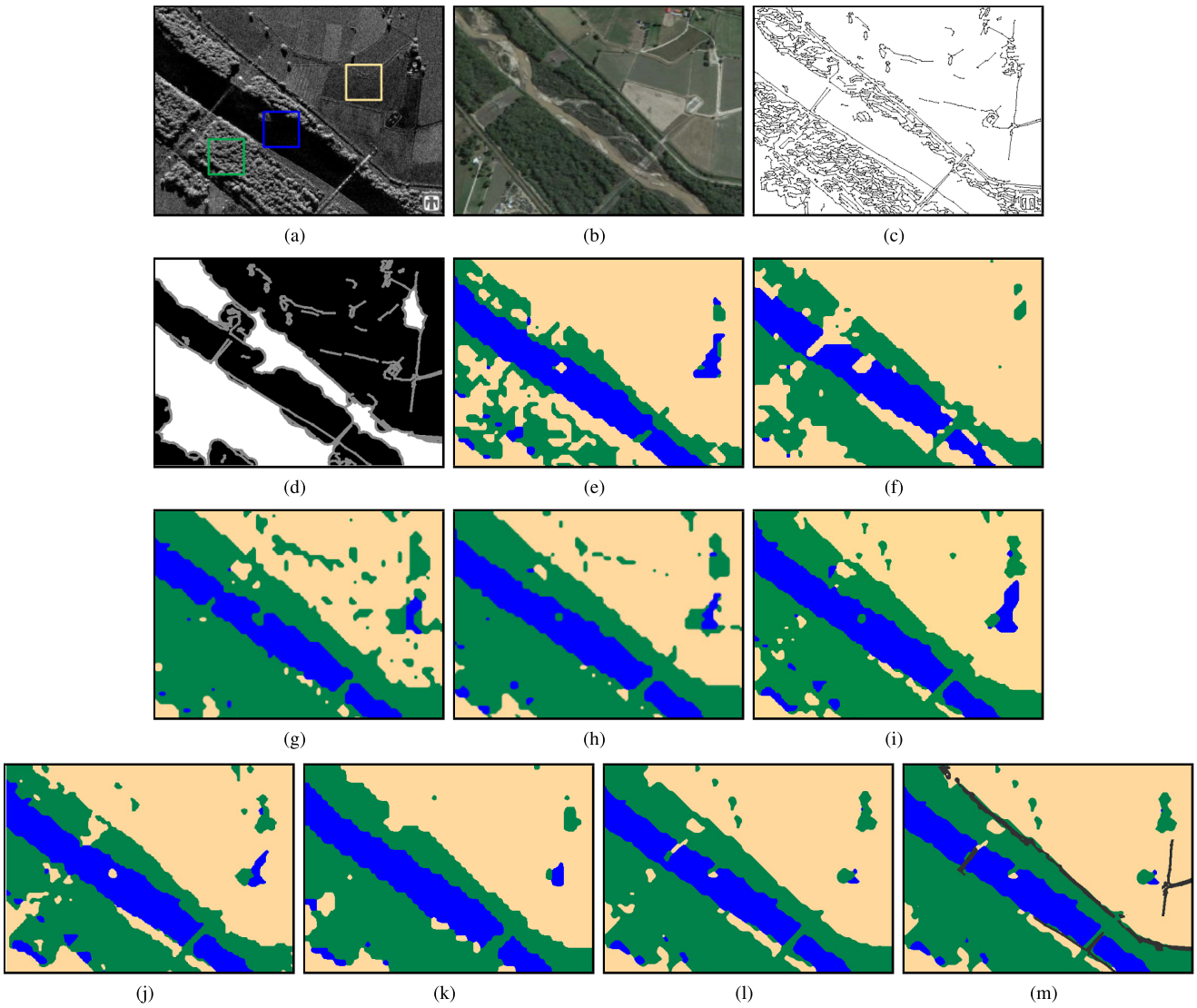


Fig. 10. Classification results of Piperiver. (a) Original image that contains three semantic scenes: forest (green), water (blue), and land (light yellow). 600 × 432, Ku-band, and 1-m resolution. (b) Corresponding optical image. (c) Sketch map, code length gain, high and low thresholds: (25, 1.2, and 0.8). (d) Regional map ($K = 7$). (e) GLCM-SVM. (f) Gabor-SVM. (g) CNN. (h) Gauss-CNN. (i) Gauss-CNN-two. (j) GCVAE-CNN. (k) GAN-CNN-two. (l) GCVAE-CNN-two. (m) Final classification result.

of the proposed method outperform the GLCM-SVM, Gabor-SVM, CNN, Gauss-CNN, Gauss-CNN-two, GCVAE-CNN, and GAN-CNN-two by 14.0%, 12.12%, 9.29%, 6.71%, 3.29%, 1.45%, and 0.54%, respectively, which sufficiently validate the effectiveness of the hybrid network with structural constraints.

E. Qualitative Analysis of Real High-Resolution SAR Images

We summarize three difficulties of SAR image scene classification: 1) water scene and shadows are very similar and hard to distinguish; 2) some scenes have unobvious interclass differences, such as residential scenes with different densities; and 3) the larger the image, the more abundant the scenes it contains. All of these make SAR image classification more difficult in large scale and complex scenes. Therefore, we choose three SAR images, including these difficulties to validate the effectiveness of the proposed method and to compare it with the seven related methods. Figs. 10(a)–12(a) are the original SAR images, where the colored squares and rectangle are the training areas that we manually labeled. Figs. 10(b)–12(b) show the corresponding optical images downloaded from Google Earth (Google, LLC, Menlo Park, CA, USA) for reference. It should be pointed out that the optical images and SAR images are not collected at the same time. Figs. 10(c)–12(c) are the sketch maps [40], and Figs. 10(d)–12(d) present the regional maps of SAR images [49]. Because the SAR images do not have ground truth, there is no standard of correct classification for quantitative analysis, only the visual comparison can be displayed in this section. In addition, all the classification results are smoothed by a 5×5 max pooling.

Although the scenes of the Piperiver image are relatively simple, the classification results of these methods are quite different, especially in the water and forest scenes. It can be seen that the regional consistency of the forest scene is poor as the result of the GLCM-SVM [Fig. 10(e)]. The classification result obtained by the Gabor-SVM [Fig. 10(f)] is smoother than that of the GLCM-SVM, but there are

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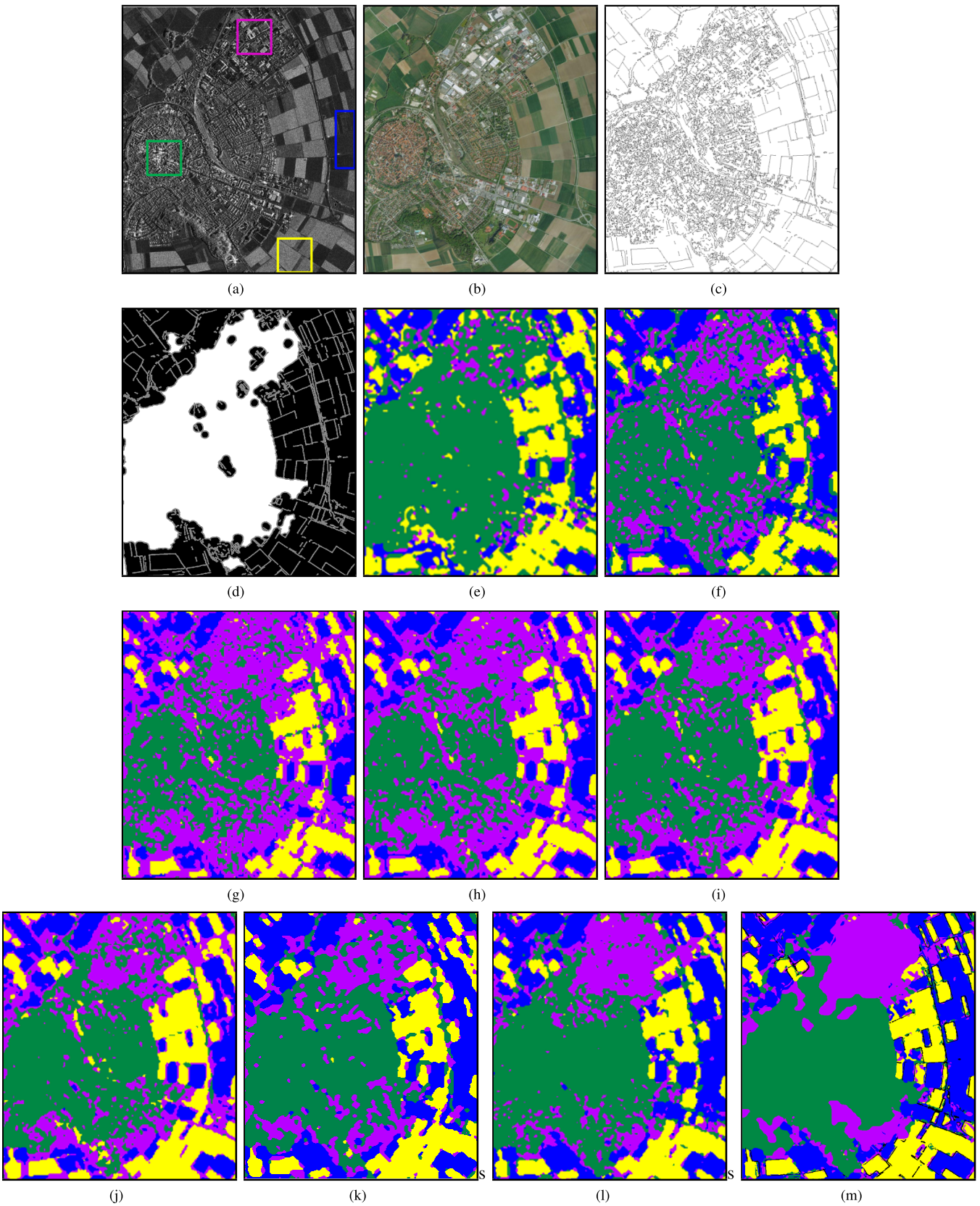


Fig. 11. Classification results of Noerdlin. (a) Original image that contains four semantic scenes: farmland (blue), industrial (magenta), residential (green), and idle-land (yellow). 1308×1452, X-band, and 1-m resolution. (b) Corresponding optical image. (c) Sketch map, code length gain, high and low thresholds: (25, 1.2, and 0.8). (d) Regional map ($K = 5$). (e) GLCM-SVM. (f) Gabor-SVM. (g) CNN. (h) Gauss-CNN. (i) Gauss-CNN-two. (j) GCVAE-CNN. (k) GAN-CNN-two. (l) GCVAE-CNN-two. (m) Final classification result.

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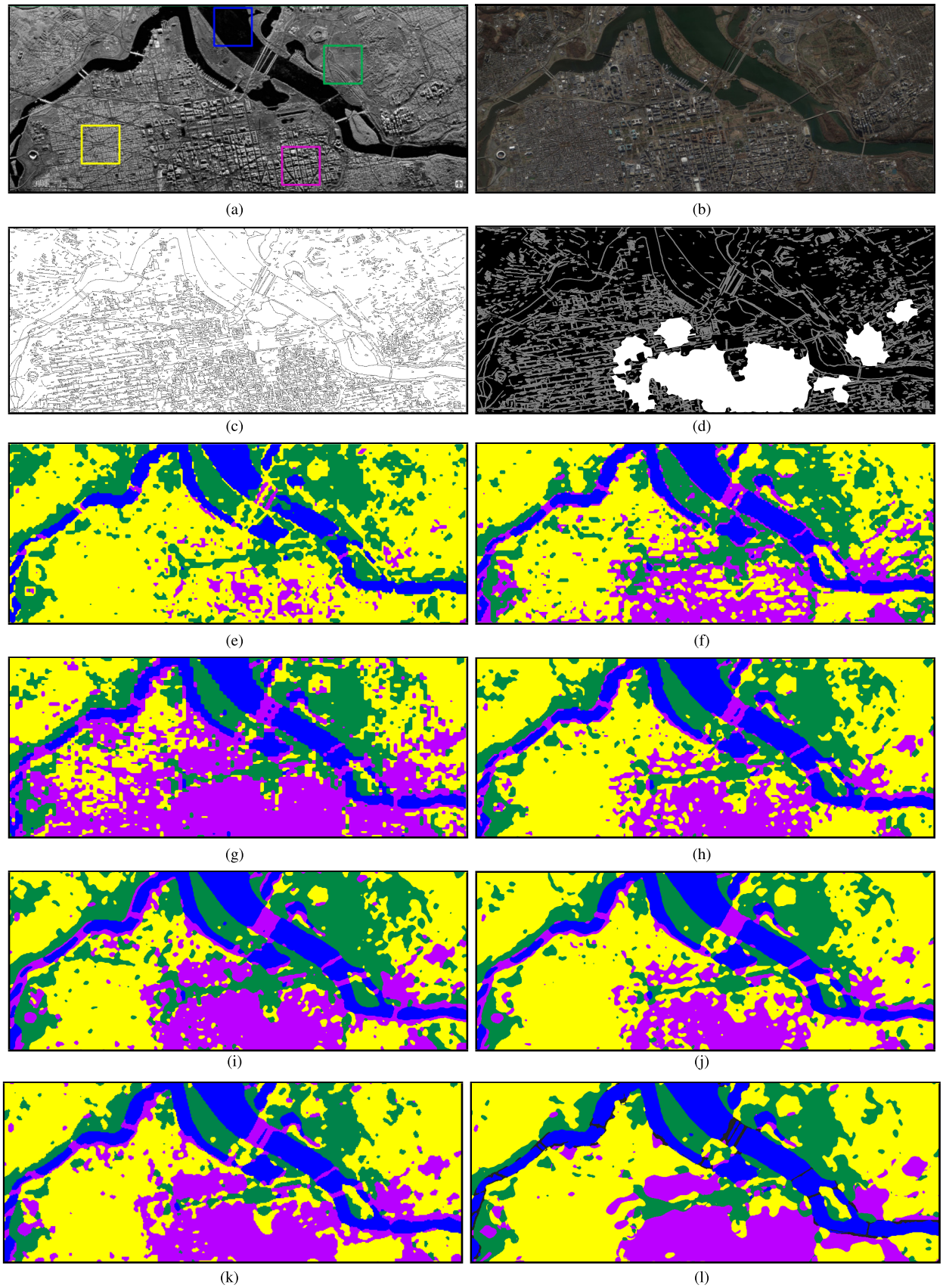
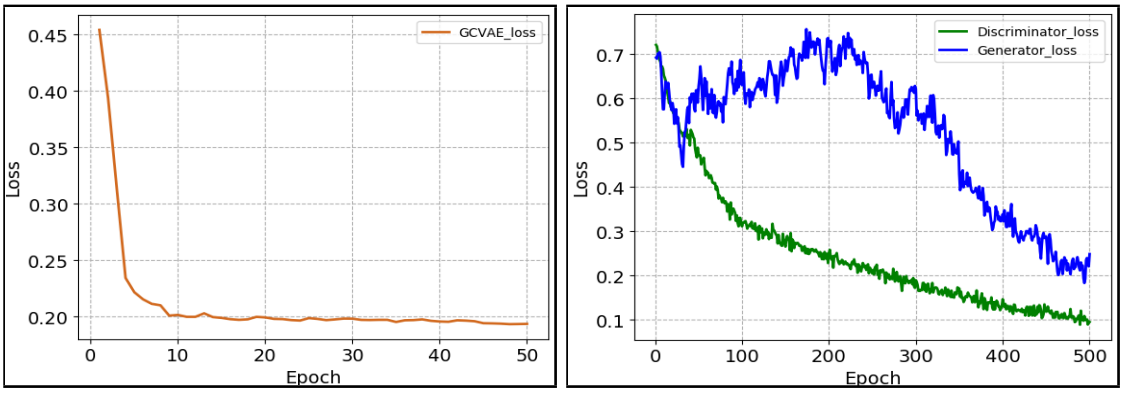


Fig. 12. Classification results of W. DC. (a) Original image that contains four semantic scenes: water (blue), land (green), sparse residential (yellow), and dense residential (magenta). 2000 × 810, Ku-band, and 1-m resolution. (b) Corresponding optical image. (c) Sketch map, code length gain, high and low thresholds: (25, 1.2, and 0.8). (d) Regional map ($K = 5$). (e) GLCM-SVM. (f) Gabor-SVM. (g) CNN. (h) Gauss-CNN. (i) Gauss-CNN-two. (j) GCVAE-CNN. (k) GAN-CNN-two. (l) Final classification result.

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(a) (b)

Fig. 13. Loss curves during training of two different networks. (a) Loss curve of GCVAE. (b) Loss curves of GAN.

some small confusing regions in the water and forest scenes. Fig. 10(g) shows the result of the CNN method, there are many misclassifications in the land scene. Because we only obtain 256 training samples on each labeled patch, it may lead to insufficient training of the parameters of the CNN. The regional consistency of the Gauss-CNN [Fig. 10(h)] and the Gauss-CNN-two [Fig. 10(i)] is better than that of the CNN. Nevertheless, some shadows are still divided into waters into these methods. Fig. 10(j) shows the result of GCVAE-CNN, and we can see that the GCVAE-CNN performs better than the results of the Gauss-CNN and Gauss-CNN-two. The result of GAN-CNN-two is shown in Fig. 10(k), which has better regional consistency than GCVAE-CNN, but the preservation of the details is less than our method [Fig. 10(l)]. Our approach realizes the best performance for SAR image scene classification and there is almost no misclassification for water scene. Fig. 10(m) shows the final classification result, in which the line targets (roads and bridges) are located to obtain a more accurate classification result [50].

Fig. 11 shows the classification results of the Noerdlin image. The GLCM-SVM [Fig. 11(e)] almost divides all industrial into the residential scene, and there are severe misclassifications. The result of the Gabor-SVM [Fig. 11(f)] obviously outperforms that of the GLCM-SVM, which indicates that the representation of features extracted by implementing the Gabor transform is superior to that employing the GLCM. The result of the CNN is presented in Fig. 11(g), it does not divide the same scene into semantically consistent regions well, and there are some over classifications in the industrial scene. This result indicates that it is not feasible to directly use the CNN for small sample learning. Fig. 11(h) and (i) shows the results of the Gauss-CNN and Gauss-CNN-two. The classification results in them are more accurate than that of the CNN, and the regional consistency of the Gauss-CNN-two method outperforms that of the Gauss-CNN, which indicates that the two constraints can retain the structural information of different terrain types to improve the classification performance. However, the Gauss-CNN-two still misclassified some lands as an industrial scene. The result of the GCVAE-CNN [Fig. 11(j)] is better than that of the Gauss-CNN. In particular, the regional consistency of the residential scene is greatly improved. Fig. 11(k) shows the result of GAN-CNN-two, it can be seen that the regional consistency of GCVAE-CNN-two [Fig. 11(l)] is slightly better than GAN-CNN-two. This is due to our approach can capture more discriminative information for SAR image scene classification. Fig. 11(m) shows the final result which improves the accuracy of boundary localization by some visual semantic rules [49], [50], and regional consistency by differential classifying of aggregated (patch size, 32×32) and homogeneous regions (24×24).

Fig. 12 shows the scene classification results of the W. DC image. We can find from Fig. 12(e) that a great number of dense residential areas are misclassified into sparse residential areas, which indicates that the GLCM features are not discriminative enough to distinguish these scenes. Fig. 12(f) shows the result of the Gabor-SVM, the regional consistency of this method is not very well, and there are some ambiguities between the dense residential and sparse residential scenes. The result of the CNN [Fig. 12(g)] is also very poor, and many sparse residential areas are mistakenly classified as dense residential areas. This means that the labeled training samples are insufficient for the CNN to train a large number of parameters, which results in the inaccurate classification result. Fig. 12(h) shows the result of the Gauss-CNN, the classification performance is significantly improved compared with that of the CNN, but there are still erroneous classifications in the sparse residential scene. Fig. 12(i) shows the classification result using the Gauss-CNN-two, which integrates two structural constraints with the Gauss-CNN, achieving better spatial connectivity. However, some waters are still misclassified into the dense residential scene. Fig. 12(j) shows the result of the GCVAE-CNN, we can find that the regional consistency of land and sparse residential scenes are improved compared with Fig. 12(i), which indicates that the GCVAE-CNN is more representative and has a stronger ability in distinguishing diverse terrains. The classification result of GAN-CNN-two is shown in Fig. 12(k), and we can see that the regional consistency of the dense residential is better than the result of the GCVAE-CNN. For better regional consistency and boundary accuracy, through information interaction among pixel space, sketch map, and regional map, we locate some line targets try our best, and classify the aggregated and

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homogeneous regions with different patch sizes (32 and 24). After that, all the results are integrated together to obtain the final classification map, as shown in Fig. 12(). It can be seen that the regional consistency of our method is better than that of the GAN-CNN-two.

There are two main reasons why the results of the proposed method are better than those of GAN-CNN-two. First, we introduce the statistical distribution information of SAR into GCVAE, which enables the network to learn more accurate distribution information of SAR images and also makes the network convergence faster in the training process. While, the GAN automatically learns the distribution of data, which is not very stable in the process of training. Their convergence performance is shown in Fig. 13. It can be seen that the convergence speed of GCVAE is more than $10\times$ faster than that of GAN. Second, the structural constraints contribute to preserving the structural information of samples, which can also make the generated samples clearer.

V. Conclusion

In this article, we have proposed a hybrid network with structural constraints for SAR image scene classification. There are three advantages of our method: 1) we add the prior information of SAR images into the SGVB model to construct a GVBI model and an unsupervised GCVAE for learning the distribution of SAR images and generating more abundant samples, which can alleviate the small-sample problem in SAR image classification. 2) To overcome the dependence of data-based methods on labeled samples, at the same time, utilize the performance advantage of data-based to make up for some limitations of model-based methods, we construct a hybrid network combining model-based with data-based methods to achieve end-to-end generation and classification. 3) The features learned by the hybrid network can capture the characteristics of distribution and structure of SAR images simultaneously, consequently improving the classification performance. However, the proposed method still has some improvements. For example, one can construct novel generation models to further improve the diversity of generated samples and design special CNNs to advance the performance of the hybrid network. In addition, how to integrate data-based and model-based methods better and complement each other is a meaningful problem worthy of further study.

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Xiaoxue Qian received the B.S. degree from the School of Electrical and Electronic Engineering, Shaanxi University of Science and Technology, Xi'an, China, in 2006, and the M.S. degree from the School of Electronic Engineering, Xidian University, Xi'an, in 2009, where she is pursuing the Ph.D. degree with the School of Artificial Intelligence.

Her research interests include machine learning, image interpretation, and pattern recognition.



Fang Liu (Senior Member, IEEE) received the B.S. degree in computer science and technology from Xi'an Jiaotong University, Xi'an, China, in 1984, and the M.S. degree in computer science and technology from Xidian University, Xi'an, in 1995.

She is a Professor with Xidian University. She has authored or coauthored over five books and over 80 articles in journals and conferences. Her research interests include image perception and pattern recognition, machine learning, evolutionary computation, and data mining.

Ms. Liu received the National Natural Science Award (second prize) in 2013.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is the Director of Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, China. He is in charge of about 40 important scientific research projects. He has authored or coauthored more than 20 monographs and a 100 articles in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is a member of the IEEE Xi'an Section Execution Committee, the Chairman of the Awards and Recognition Committee, the Vice Board Chairperson of the Chinese Association of Artificial Intelligence, a Councilor of the Chinese Institute of Electronics, a Committee Member of the Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.



Xiangrong Zhang (Senior Member, IEEE) received the B.S. and M.S. degrees in computer science and the Ph.D. degree in electronic engineering from Xidian University, Xi'an, China, in 1999, 2003, and 2006, respectively.

She has been a Visiting Scientist with the Laboratory of Computer Science and Artificial Intelligence, Massachusetts Institute of Technology, Cambridge, MA, USA, since 2015. She is a Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. Her research interests include pattern recognition, machine learning, and image analysis and understanding.

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Puhua Chen (Member, IEEE) received the B.S. degree in environmental engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2009, and the Ph.D. degree from Xidian University, Xi'an, China, in 2016.

She is a Post-Doctoral Researcher with the School of Computer Science and Technology, Xidian University. Her current research interests include machine learning, pattern recognition, and synthetic aperture radar image interpretation.



Lingling Li (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2017, respectively.

From 2013 to 2014, she was an Exchange Ph.D. Student with the Intelligent Systems Group, Department of Computer Science and Artificial Intelligence, University of the Basque Country UPV/EHU, Spain. She is a Lecturer with the School of Artificial Intelligence, Xidian University. Her research interests include image processing, deep learning, and pattern recognition.



Jing Gu (Member, IEEE) received the B.S. and M.S. degrees from the Xi'an University of Technology, Xi'an, China, in 2007 and 2010, respectively, and the Ph.D. degree from Xidian University, Xi'an, in 2016.

She is a Post-Doctoral Researcher with the School of Computer Science and Technology, Xidian University. Her research interests include image processing, machine learning, and pattern recognition.



Yuanhao Cui received the B.S. degree from the Changchun University of Science and Technology, Changchun, China, in 2015. He is pursuing the Ph.D. degree in computer application technology with Xidian University, Xi'an, China.

His main research interests include big data perception and pattern recognition, machine learning, and intelligent image processing.

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